Code-Layout, Readability and Reusability

Date	30 June 2026
Team ID	XXXXXX
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Code-Layout, Readability and Reusability:

This document evaluates the quality of the codebase in terms of its structure, readability, and reusability. Well-organized code with proper naming conventions, comments, and modular design ensures that the project is maintainable and can be extended or reused in future development.

Code Layout Checklist:

S.No	Code Quality Parameter	Description	Followed (Yes/No/Partial)	Remarks
1	Consistent Indentation	Code is properly indented to improve readability and maintain a uniform coding style.	Yes	Proper indentation is maintained throughout the project.
2	Proper File Structure	Project files are organized into separate folders for templates, static files, models, datasets, and utilities.	Yes	Project follows a modular folder structure.
3	Meaningful Variable Names	Variables and functions use clear and descriptive names for better understanding.	Yes	Names such as nitrogen, temperature, and predict_crop are used.
4	Function / Method Names	Functions are named according to their purpose and follow standard naming conventions.	Yes	Functions like predict_crop() and preprocessing methods are clearly named.
5	Code Comments	Important sections of the code include comments to explain their functionality.	Yes	Comments are added where necessary for clarity.
6	Modular Design	The application is divided into reusable modules such as preprocessing, prediction, frontend, and backend.	Yes	Backend, frontend, and ML modules are separated.
7	No Redundant Code	Duplicate or unnecessary code is minimized to improve maintainability.	Yes	Repeated code has been avoided.
8	Error Handling	The application validates inputs and handles invalid data gracefully.	Yes	Input validation is implemented before prediction.

Reusable Components / Modules:

S.No	Component / Module Name	Language / Technology	Where Reused	Reusability Level
1	Crop Prediction Model	Python, Scikit-learn	Prediction Module	High
2	Input Validation	Python	Form Validation	High
3	Flask Routes	Flask	Navigation & Prediction	High
4	HTML Templates	HTML, Bootstrap	Home, About, FindYourCrop	High
5	CSS Styles	CSS	All Web Pages	Medium

Overall Code Quality Assessment:

Aspect	Rating (1–5)	Comments
Code Layout & Structure	5	Well organized project structure.
Readability	5	Easy to understand and maintain.
Reusability	5	Modules can be reused in future projects.
Documentation / Comments	4	Important code sections are documented.
Overall Score	4.8 / 5	Clean, modular, and maintainable code.